Bonus 4: A Rectangle Rotated Around the z Axis, in 3D

Week 4 — PNG submission

Submission: submit a single PNG file on the course Moodle page.

Optional bonus: a rectangle rotated around the z axis

Take a vertical rectangle in the plane $x = 0$ with corners $(0, 0, 0.8)$, $(0, 0, 1.8)$, $(0, 1, 1.8)$, $(0, 1, 0.8)$. Rotate it around the z-axis by every angle $\theta = 0°, 30°, \ldots, 330°$, each time applying

$$R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

to every vertex, and draw all rotated copies together as a 3D "fan", with the z-axis drawn as a line. Save the result as `week04-rectangle-3d.png`.

The starting point is the script `bonus-week04-rectangle-3d.jl` in the Week 4 materials folder.

How to plot in 3D

A 3D plot works just like a 2D plot, except every point has three coordinates, so you pass three vectors of equal length: the x coordinates, the y coordinates, and the z coordinates. The basic building block is `plot(xs, ys, zs; seriestype = :path3d)`, which draws a polyline through the points:

```
xs = [0.0, 1.0, 1.0, 0.0, 0.0]
ys = [0.0, 0.0, 1.0, 1.0, 0.0]
zs = [0.0, 0.0, 0.0, 0.0, 1.0]
plot(xs, ys, zs; seriestype = :path3d)
```

A rectangle is a closed polygon: list its vertices, then append the first vertex again so the line returns to its start. `plot` creates a new figure; `plot!` adds a series to an existing one. That is all the bonus needs:

- loop over the angles $\theta = 0°, 30°, \ldots, 330°$;
- for each angle, rotate the four vertices with $R_z(\theta)$ and build the closed `xs`, `ys`, `zs` loops (use `push!` in a loop);
- add the rectangle with `plot!(p, xs, ys, zs; seriestype = :path3d, lw = 2)`;
- draw the z-axis as a second series: `plot!(p, [0,0], [0,0], [-0.4, 3.0]; seriestype = :path3d, color = :black, lw = 1.5)`.

For a stable, comparable figure set the title, the three axis labels, `legend = false`, `camera = (30, 30)`, and `size = (800, 600)` in the initial `plot` call. Display with `display(p)` and save exactly the same figure with `savefig(p, "week04-rectangle-3d.png")`.

PNG requirements (automatic check)

The submitted `week04-rectangle-3d.png` is compared automatically with a reference image, so nothing about the rendering may change:

- the point of view (`camera=(30,30)`), the figure size (`size=(800,600)`), and the resolution;
- the line width of the rotated rectangles (`lw=2`) and of the axis line (`lw=1.5`);
- the title, the axis labels, the axis limits, `legend=false`, and the geometry of the rectangle and the axis.

Use `savefig(p, "week04-rectangle-3d.png")` after the plot so the saved PNG has exactly these settings.

Submission

Submit the single PNG file `week04-rectangle-3d.png` on the course Moodle page.

Checking

Check one rotation by hand: under $R_z(90^\circ)$ the corner $(0, 1, 0.8)$ moves to $(-1, 0, 0.8)$, and the z coordinate of every vertex never changes. If your saved PNG was produced without changing any visual setting, it should pass the automatic comparison.